

Development of an Intelligent System for Evaluating Risk Factors Affecting Public Transport Drivers

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Abstract— Current state of public reliance has raised concerns about the safety of public transport drivers. Public transport drivers exposed to several risks such as drowsiness, drunk driving and health issues. This study proposes an intelligent system for evaluating risk factors affecting public transport drivers. The proposed system consist of three main components: (1) Drowsiness detection with YOLO based algorithms, which analyzes face signatures for the fatigues; (2) Alcohol intoxication detection, using MQ-3 gas sensor on Raspberry PI based mini computer which will detect alcohol level before vehicle ignition; (3) Heartbeat monitoring using a heartbeat sensor and an MCP3008 analog converter to track drivers heartbeat continuously. The goal is to create real time monitoring system using embedded hardware and machine learning. Current results shows that YOLOv5 model significantly outperformed other models in drowsiness detection problem.

Keywords—Machine learning; YOLO; MQ-3; Raspberry PI; Public Transportation; video processing; heartbeat monitoring;

I. INTRODUCTION

Urbanisation with public transport reliance has a significant impact in the world we live in today. Public transport drivers are exposed to several risk factors, such as drowsiness, health conditions and intoxication during driving. Studies show that these factors are crucial since they affect drivers performance, reaction times and overall road safety [1]. In this article, we offer a solution as an

intelligent system for evaluating such risk factors of public transport drivers.

One of the most serious problems related to the safety of a car driver on public transport is drowsiness, which significantly impairs cognitive function and reaction time. Similarly, Internet of Things-based solutions have been proposed to detect alcohol consumption and prevent driving violations. Such systems use Raspberry pi and its gas sensors such as MQ-3 that can measure alcohol concentration in the air and determine the sobriety level of the driver before operating the vehicle [2] [3].

In addition, an assessment of the driver's health parameters, such as heart rate and body temperature, can provide insight into overall fitness for duty. Several researches show that integrating min-computer based health monitoring systems that have continuous biometric tracking can reduce health risks while driving [4] [5].

Given the complexity of the risk factors affecting public transport drivers, a comprehensive and intelligent assessment system is needed. The novelty of this project is the integrated approach of three different modules, real-time drowsiness detection using YOLO-based models, pre-ignition alcohol sensing, and continuous heartbeat monitoring a combination not previously implemented in a unified, low-cost embedded system for public transport safety

II. LITERATURE REVIEW

The literature on intelligent risk assessment for public transport drivers covers multiple subjects, including

drowsiness detection, impaired driving prevention, real-time object detection, and health monitoring.

Research on sleepiness detection is primarily based on visual analysis of behavior and physiological indicators. Even though “drowsiness” is the more commonly used term, “fatigue” is also frequently employed. Despite their meanings, the terms fatigue and drowsiness are often used interchangeably [6]. Fatigue refers to an unwillingness or inability to continue a task due to physical or mental exhaustion or prolonged engagement in repetitive tasks [7]. In contrast, sleepiness or drowsiness detection describes the irresistible urge to fall asleep, driven primarily by a biological need for rest.

Study by Shivani et al. and Kumar and Patra shows the effectiveness of machine learning algorithms that can detect drowsiness level based on eye-closing patterns, facial landmarks and yawning detection [1]. Furthermore, Borra et al. propose a real-time driver monitoring system that combines behavioral and physiological data to improve detection accuracy [8]. These studies highlight the importance of advanced sleepiness detection algorithms in such monitoring systems.

Driving under the influence of alcohol can lead to severe road accidents. According to Simmons et al. Alcohol consumption increases hazard chance, slows reaction time and worsens motor coordination [9]. Research has demonstrated correlation between blood alcohol concentration (BAC) level and driving impairment. Even low BAC levels (around 0.04-0.06%) can significantly increase crash rates, delay hazard responses, and reduce lateral vehicle control, as evidenced by increased lane variability and excursions [9].

To address this issue, IoT-based systems for alcohol detection have emerged as promising solutions. Studies by Viswanatha et al. and McShane et al. Show the efficiency of Raspberry Pi based systems that use gas sensors to measure alcohol concentration in human breath [2] [10]. Moreover, this system fires real time alerts and has vehicle control to prevent drunk driving.

There are several models that are particularly effective on object detection. YOLOv5 can be considered as one of the effective models amongst them. Chandana & Ramacahandara and Liu et al. researches the application of YOLOv5 networks in real-time traffic monitoring, obstacle detection and pedestrian recognition [11]. Furthermore, Karthi et al. evaluated YOLOv5 across various benchmarks and custom datasets, demonstrating its robustness in detecting multiple object classes simultaneously [12]. Small size paired with implementation in PyTorch simplifies its deployment on mobile devices and minicomputers. Applications of YOLOv5 include real-time traffic monitoring, pedestrian recognition, and obstacle detection, significantly enhancing situational awareness and supporting proactive safety measures.

It is obvious that the well-being of drivers is the number one priority for public transport accident

prevention. Studies by Sollu et al. and Pardeshi et al. examine the integration of health monitoring systems [4]. As for data collection Raspberry PI based mini computers are common choices for such systems. Such systems provide continuous monitoring of heartbeat, body temperature and other vital parameters.

In addition to technical developments in driver monitoring systems, recent pedagogical studies have highlighted the importance of integrating robotic technologies into educational programs. For instance, Serik and Nurgaliyeva [13] examined the enhancement of future computer science teachers' competence through the development of mobile robots using Arduino, Lego Mindstorms EV3, and machine learning technologies. Their findings emphasized the significance of combining theoretical and practical components in the training process, which parallels the multidisciplinary approach of intelligent risk evaluation systems for public transportation drivers.

Research by Arakawa shows the importance of heartbeat monitoring systems in automotive applications, highlighting both wearable and non-wearable technologies. Wearable devices, including wristwatch-type, ring-type, and shirt-type monitors, offer unobtrusive continuous measurement of the driver's biometric data. Non-wearable systems integrated into vehicle components, such as steering wheels, seats, seat belts, and portable devices, provide alternative methods for monitoring without directly affecting driver comfort. Arakawa notes that accurate heartbeat detection can indicate the driver's arousal and fatigue levels, critical indicators of potential health issues or diminished driving competency. Such systems can help detect sudden health changes in drivers, enabling timely intervention and accident prevention [14].

Existing literature shows the importance of including risk assessment components such as drowsiness detection, alcohol level detection and health monitoring into public transports. Proposed system builds upon these advancements to develop a comprehensive intelligent evaluation system aimed at ensuring the safety and efficiency of public transportation.

III. PROPOSED SYSTEM

Proposed system contains a web camera for drowsiness detection. Under the hood the camera will transfer real-time video to the mini computer. Then a YOLO-based object detection model will identify signs of drowsiness. By analyzing key facial features, such as eye closure patterns and yawning, the system will classify the driver's alertness status. If the classified level of drowsiness is above the risk threshold then an appropriate sound alert will be fired from the mini-computer. Figure 1 demonstrates overall flow for sleepiness detection.

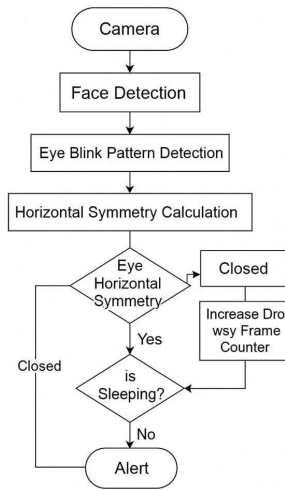


Figure 1. Drowsiness detection flow

The module for identifying drowsiness will be implemented via YOLO based algorithms of object detection. Since original YOLOv5 algorithms were designed primarily for object detection it might be challenging to detect small features such as closed eyes on a human's face [11]. However, all of these can be negotiated with the quality dataset containing all edge cases and all stages of drowsiness.

During implementation of the “drowsiness” detection part of the system two different methods were considered for the usage.

The first method is about using numpy and detecting the area around the eyes that will fire an alert if the radius is less than the given threshold. Figure 2 demonstrates the coefficient of radius during the awake state of the eyes.



Figure 2. Awake state of eyes with first method

Furthermore, the method was tested with closed eyes. After detecting the coefficient of closed eyes, the program decides to fire an alarm. Figure 3 demonstrates the coefficient of radius during the closed state of the eyes.

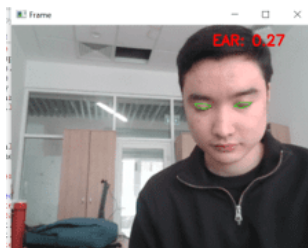


Figure 3. Closed state of eyes with second method

Next test case was to check how well the program would work with obstacles around eyes like glasses. Figure 4 demonstrates coefficient of radius with glasses on.

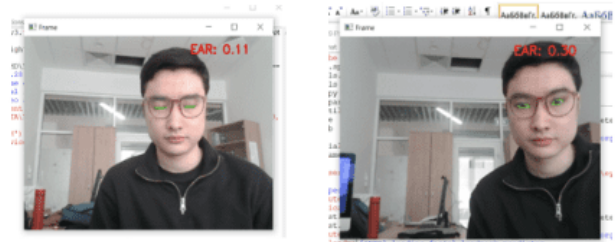


Figure 4. Closed state of eyes with second method

Second method was to train the YOLOv5 based model in order to detect drowsiness. Figure 5 demonstrates how the model detects awake state.

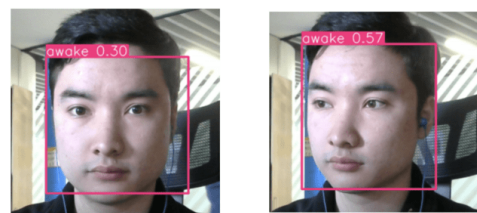


Figure 5. Awake state with second method

With the awake state model tested on sleep signs. The model detects yawning, closed eyes and tilted head as signs of drowsiness. Figure 6 demonstrates the sleepy state of the second model.

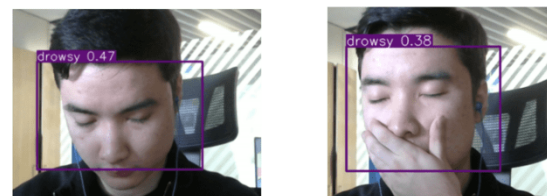


Figure 6. Drowsy state with second method

The YOLO based model provides more precise prediction with complex gesture recognition such as yawning and head tilting. Hence, it is decided to use the YOLOv5 model as the base model for the drowsiness detection part of a system.

Dataset used for YOLOv5 based drowsiness detection component contains several images and video recordings of public transport drivers captured during regular operational hours. Dataset covers wide area of conditions indicating driver's fatigue. That includes different stages of eye closure, yawning episodes, and head nodding events. Moreover, it covers diverse lighting conditions, driver postures, facial orientations, and occlusion scenarios to ensure reliability against real world scenarios.

The first stage of a proportion is to gather a large dataset consisting of images and videos of drivers during their work hours. This dataset is intended to contain

various states of fatigues of a driver. States will cover all aspects of drowsiness, including eyes closed, yawning, and even head nodding.

The accuracy of the model is evaluated using the average system response time to prolonged eye closure events. Test results are based on real-world scenario data, where every millisecond is crucial for the system to react promptly and prevent potential hazards.

Figure 7 illustrates formula of Eye Aspect Ratio (EAR):

$$EAR = (\|P_2 - P_6\| + \|P_3 - P_5\|) / (2 * \|P_1 - P_4\|)$$

Where P_1 to P_6 demonstrates 2D landmark coordinates of the eye. $\|.\|$ denotes the Euclidian distance between two points. The numerator represents vertical eye distances and denominator represents the horizontal eye width.

Table 1 demonstrates comparison table between these two methods

TABLE I. COMPARISON BETWEEN TWO METHODS

Method	AVG Detection speed (s)	EAR Integration	Key Features
Numpy-based EAR	0.07	Direct	Basic eye landmark analysis, no gesture recognition
YOLOv5	0.03	Indirect	Real-time detection, pre-trained COCO weights, moderate accuracy

The proposed YOLOv5-based model was compared against existing solutions that used traditional methods such as the Eye Aspect Ratio (EAR) calculation. Figure 7 demonstrates close lookup of the eye with points for the EAR calculation.

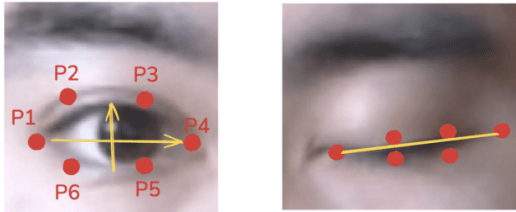


Figure 7. Close lookup for the EAR calculation

YOLOv5 provided significantly faster detection speeds, averaging 0.03 seconds per frame compared to 0.07 with numpy-based EAR methods.

Accurate gesture recognition of fatigue signs such as yawning and head tilting was observed with the YOLOv5 model. Thus, validating suitability for comprehensive drowsiness detection.

Even though EAR integration in YOLOv5 model was indirect the model's adaptability and precision in real world operational conditions underscored its effectiveness compared to simpler eye landmark analyses.

Another comparison was made between YOLOv5 model and MobileNetV2 for drowsiness detection. MobileNetV2 is a convolutional neural network (CNN) designed specifically for mobile and edge device connections [15-16]. For the drowsiness detection MobileNetV2 model was fine-tuned using transfer learning. Table 2 demonstrates comparison between YOLOv5 metrics and MobileNetV2 metrics.

TABLE II. YOLOV5 VS MOBILENETV2 METRICS

Algorithm	Accuracy	Precision	Recall	F1 score
YOLOv5	0.905	0.929	0.928	0.928
MobileNetV2	0.476	0.636	0.500	0.560

When it comes to drowsiness detection YOLOv5 outperformed MobileNetV2 in all aspects. Having almost twice as much greater performance than MobileNetV2 YOLOv5 comes as the leader [17-18]. Thus, we can conclude that even though MobileNetV2 offers computational efficiency it lacks in classification problem compared to YOLOv5

YOLOv5 demonstrates high precision in detecting awake and drowsy states according to the given classification problem. However, it fails to identify background image class by misclassifying it entirely as either awake or drowsy. This could result in a false positive in real-world applications where background frame could consist of empty seats or no driver. To mitigate it model might need better representation of background class with more training data or different loss function that responsible for class imbalance [19-20]. Figure 8 shows confusion matrix of the YOLOv5 model.

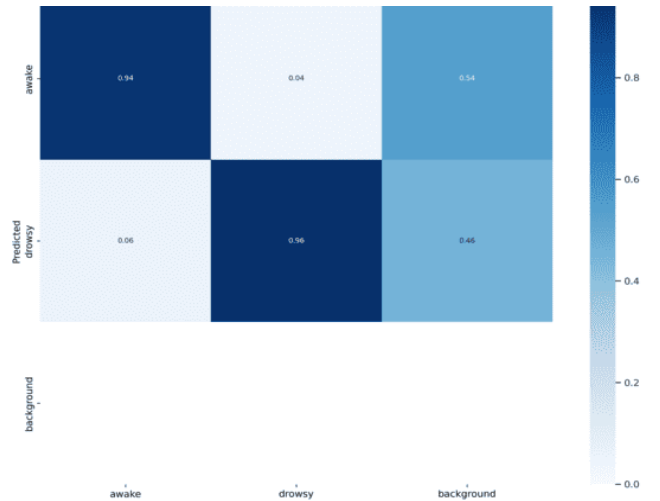


Figure 8. Confusion Matrix of the YOLOv5 model

During development phase of first component several challenges were encountered:

1) Small feature detection

At first, YOLOv5 was designed for larger objects. Hence, there was initial struggle with accurately detecting small facial features such as partially closed eyes. This issue

was mitigated by increasing the dataset size and fine-tuning model.

2) Occlusion Handling

Accurate detection in case of drivers wearing glasses or hats required additional commented examples and specialized trainings to reduce false positives and negatives.

3) Real time performance

Achieving real time response with minimal response time was another challenge for the first part of the system. Techniques such as network pruning and model quantization were explored to maintain accuracy while reducing computational load.

Next component of the proposed system is alcohol detection. Alcohol detection will be implemented via the MQ-3 sensor of Raspberry PI based machines. This module will determine the concentration of alcohol from the driver's breath before ignition of the vehicle. If the alcohol concentration exceeds a predefined threshold, the system will generate an alert and prevent vehicle operation.

Module itself will consist of four different components connected to Raspberry PI pico based mini-computer. MQ-3 alcohol sensor will be connected to ADC 26th pin. Engine connected to the 10th pin, while the alarm is connected to the 11th pin. Finally, a power supply of 5 volts will be connected to the system. Figure 9 demonstrates the connection schema of a mini-computer.

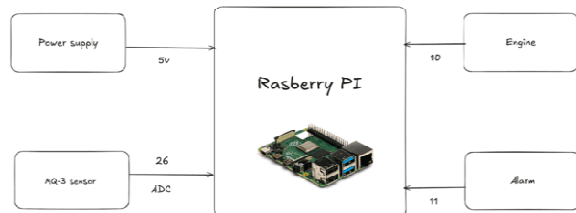


Figure 9. Alcohol detection module block schema

The MQ-3 sensor must be calibrated to guarantee accuracy. Which can be accomplished by adjusting sensitivity of alcohol concentration accordingly. Sensor logic can be programmed via RPI.GPIO library in python for pin controls and ADC module data extraction. A proposed script should continuously fetch sensor readings then apply conversion formulas to change the data, and check the threshold for alarm in real time.

Last part of the solution is a heartbeat monitoring system. Proposed system will use a heartbeat sensor integrated in a raspberry pi machine. Heartbeat data will be given upon placing a finger on that sensor. Preliminary placement of a sensor is a wheel in a public transport.

Hardware setup will consist of integrating MCP3008 chip that will help ADC to convert analog signals from heartbeat sensors into digital values of Raspberry PI. Mini-computer in themselves lack built-in ADC functionality, hence integrating additional MCP3008 chips is important. Several jumper wires are used to connect

MCP3008 with a mini-computer. VDD and VREF connected to 3.3V power. AGND and DGND ground connections as well as CLK (Clock) pins. Additionally, DOUT (Data Out) and DIN (Data In) are used for SPI communication. Figure 10 shows the full hardware scheme of a system.

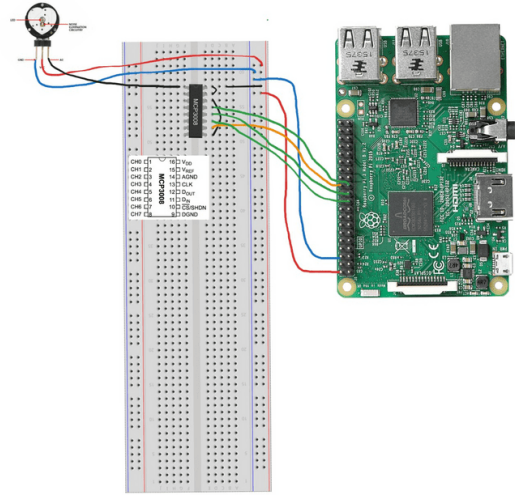


Figure 10. Heartbeat monitoring hardware schema

Sensors will monitor heartbeat by identifying variations of blood flow through the finger within each pulse of a body. As for the normal, boundaries of a heartbeat is considered to be 60 to 100 bpm. When a heartbeat is not covered within the described boundaries, an alarm signal will be sent with the alarm module.

IV. CONCLUSION

The proposed intelligent system combines several modules for risk assessment. Using advanced techniques of machine learning and computer vision combined with IoT technologies, the system identifies several risk factors. Those factors include drowsiness detection, alcohol intoxication detection and heartbeat monitoring.

Drowsiness detection module uses YOLOv5 based algorithms in order to evaluate the sleepiness level of the driver. The use of YOLOv5 provides high accuracy in identifying drowsiness related features such as eye closure, yawning and head tilting. Furthermore, YOLOv5 outperformed other model MobileNetV2 and NumPy based drowsiness detection system based on EAR score. Web camera placed facing the driver will send real time video to a Raspberry PI based mini computer that will evaluate the picture based on a provided dataset.

As for the alcohol detection module, it will be implemented via MQ-3 sensor and drivers will be tested before vehicle ignition, preventing impaired driving. Sensor itself will be connected to Raspberry Pi Pico based minicomputer. The MQ-3 sensor is connected via analog input (ADC 26th pin) and is configured to detect alcohol level above defined threshold. If the threshold is reached, the system activates an alert and disables the engine to prevent impaired driving.

The system will monitor heartbeat using MCP3008-integrated, heartbeat sensor, to access physical condition in real time. Sensor itself will be embedded within the steering wheel will allow to capture pulse signals when the driver places their finger on the sensor. However, Raspberry Pi board by itself lacks built-in analog-to-digital converter MCP3008 ADC chip will be used in the setup. The integration of MCP3008 chip enables accurate conversion of heartbeat to digital signals, making it possible for embedded applications.

This versatile approach combining IoT, machine learning and embedded systems, enhances roads safety and sets the foundation for future innovations. However, future research is required to improve small feature detection, reduce false positive and fix background occlusions. Moreover, future work may include integration with biometric parameters such as temperature, blood oxygen level, as well as expanding training dataset to include more drivers and realistic video and images.

Combination of such approaches can ensure proactive engagement, reducing likelihood of accidents and improving road safety.

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